



Brand X

A young executive of a modest-sized chemical company—call it Brand X—was trying to make sense of the pricing of one of its products, call it chemical Z. She had before her data for the last 40 quarters (10 years).¹ Why was the price jumping around so much quarter to quarter? These and other questions led her to call you for advice about pricing.

Chemical Z

Chemical Z is a proprietary product of Brand X. Brand X protects its intellectual property through a series of patents on the compound itself and the processes by which it is produced. In addition, some aspects of the production process are a trade secret. Consequently, Brand X faces no direct competition in the market for Chemical Z.

Chemical Z is purchased by other manufacturers. Over the last ten years, all indications have been that demand has been fairly stable. As one employee noted, “this quarter we charged \$3.06 a gallon and sold 4.2 million gallons; three years earlier we charged \$3.07 a gallon and sold 4.2 million gallons.”

Chemical Z is made in a plant dedicated to its production. The plant was built a little over ten years ago at a cost of \$360 million for the plant and the equipment inside. Because both have a 30-year life expectancy, the company is depreciating the plant and equipment over 30 years. Its method of depreciation is straight-line. It is, therefore, taking a charge of \$3 million per quarter for plant and equipment. The nature of Chemical Z production is such that the plant is not suitable for other manufacturing.

Production requires twenty hourly employees. Currently, each hourly employee receives \$29.19 per hour. The plant can produce 10,000 gallons of Chemical Z per hour of operation.

Costs and Pricing

Table 1 provides data on costs and the price per gallon for Chemical Z for Quarter 40 (the last quarter).

Questioning reveals that Brand X has always used the formulæ employed in Table 1 to calculate both unit cost and price. Questioning further reveals that the volatility in Brand X’s price is primarily due to volatility in the price of the raw materials. Brand X locks in the price for the raw materials at the beginning of each quarter (*e.g.*, it locked in \$1.69/gallon at the beginning of

¹You have a copy of these data too; they’re in the file *chemical_plant.xls*.

Quarter 40) and then fixes the price it charges its customers shortly thereafter. Wages, salaries, and general overhead have also changed over time, but their change has been a fairly predictable 0.2% increase per quarter.

Your Assignment

The executive who has hired you wants you to advise her on how Brand X should set prices going forward. “Plus we’d be happy for any other advice you can supply about our Chemical-Z operations,” she remarks.

One caution though: “Our CEO is really old school. He doesn’t think much of MBA programs and likes to run the firm by various rules of thumb,” the executive tells you. “So to convince him to adopt your advice, you’ll need to demonstrate that your pricing method would have yielded greater profits over the previous 40 quarters.”

Table 1: Quarter 40 Costs for Chemical Z Plant

Item	Cost ^a
Salaried Employees	\$216,208
Hourly Employees	\$246,663
Total Labor costs	\$462,871
Raw Materials	\$7,140,948
General Overhead ^b	\$216,208
Depreciation	\$3,000,000
Total Overhead	\$3,216,208
Total Costs	\$10,820,026
Gallons sold previous qtr.	4,259,357
Salary labor/gal. ^c	5¢
Hourly labor/gal. ^d	6¢
Labor/gal.	11¢
Raw materials/gal. ^e	\$1.69
Gen'l O/H/gal. ^f	5¢
Depreciation/gal. ^f	70¢
O/H/gal.	75¢
Unit Cost	\$2.55
Price/gal. ^g	\$3.06
Gallons sold	4,225,413
Revenue	\$12,929,764
Profit	\$2,109,737

^aDue to rounding, not all items may sum to the total shown.

^bThis includes maintenance, insurance, spare parts, office supplies, etc.

^cCalculated as the cost of salaried employees divided by the previous quarter's output.

^dTwenty employees times the hourly wage divided by 10,000.

^eMarket price.

^fCalculated as overhead divided by previous quarter's output.

^gSet at 20% above unit cost.